

CONFIDENCE INTERVALS

STATISTICAL INFERENCE

One of the powerful uses of statistics: statistical inference about population parameters using sample data. We use data from a sample to make conclusions about a population.

Inference methods help us to predict how close a sample statistic falls to the population parameter. We can make decisions and predictions about populations even if we have data for relatively few subjects from that population.

There are two types of statistical inference methods— confidence intervals for population parameters and testing hypotheses about particular parameter values.

CONFIDENCE INTERVALS: DEFINITION

Suppose that we have an independent and identically distributed (iid) sample of data $X_1, \dots, X_n$, where $X_i \stackrel{iid}{\sim} F$, from some distribution function F . Furthermore, suppose that the distribution F depends on some parameter θ and we have formed an estimate of the parameter (statistic), denoted $\hat{\theta} = s(\mathbf{X})$, where $\mathbf{X} = (X_1, \dots, X_n)^T$ is the sample of data. As a reminder, θ , the parameter, is a function of the distribution and $\hat{\theta}$ is a function of the sample.

Given a confidence level $1 - \alpha \in (0, 1)$, the probabilistic statement:

$$P(a(\hat{\theta}) < \theta < b(\hat{\theta})) = 1 - \alpha$$

defines a $100(1 - \alpha)\%$ confidence interval for the unknown parameter θ . The confidence interval endpoints, $a(\cdot)$ and $b(\cdot)$, are functions of the statistic $\hat{\theta}$, which is a random variable. Thus, the $100(1 - \alpha)\%$ confidence interval provides a range of values depending on $\hat{\theta}$ such that the probability of θ being within the interval is $1 - \alpha$.

CONFIDENCE INTERVALS EXAMPLES

Confidence intervals with **mathematical models**.

- Example 1: Confidence interval for μ when the data comes from a Normal distribution. This is a classical example that you should know from your basic statistics course. It will be developed on the blackboard.
- Example 2: Confidence interval for σ^2 when the data comes from a Normal distribution.

AN EXAMPLE WITH NUMBERS: NUMBER OF HOURS SPENT WATCHING TV

A confidence interval for a population mean:

How much of the typical person's day is spent in front of the TV?

Excessive TV watching has been named as one factor, other than diet, for the increasing proportion of obese Americans. A recent General Social Survey asked respondents, *On the average day, about how many hours do you personally watch television?*

The sample size was 1324, the sample mean was 2.98, the sample standard deviation was 2.66, the standard error of the sample mean was 0.0731, and the 95 % confidence interval for the population mean time μ spent watching TV goes from 2.84 to 3.12 hours per day.

CONFIDENCE INTERVALS: INTERPRETATION

Unfortunately, confidence intervals don't have the interpretation that one might expect (or hope) for ... and are often misinterpreted. Incorrect interpretations of CIs are also prevalent in scientific papers.

The **most common misinterpretation** of CIs is that there is a $100(1 - \alpha) \%$ chance that $a(\hat{\theta}) < \theta < b(\hat{\theta})$ for a given estimate $\hat{\theta}$. This is incorrect because for any given estimate $\hat{\theta}$ and corresponding interval $(a(\hat{\theta}), b(\hat{\theta}))$, the inequality statement is either **true** or **false**.

CONFIDENCE INTERVALS: INTERPRETATION

The **correct interpretation** of a confidence interval is as follows: suppose that we repeat our experiment a large number of independent times, i.e., we collect R independent samples of data, each of size n . Let $\hat{\theta}_r$ denote the estimate of θ for the r -th sample of data, and let $(a(\hat{\theta}_r), b(\hat{\theta}_r))$ denote the confidence interval formed from $\hat{\theta}_r$, $r = 1, \dots, R$. As the number of replications $R \rightarrow \infty$:

$$\frac{1}{R} \sum_{r=1}^R I(a(\hat{\theta}_r) < \theta < b(\hat{\theta}_r)) \rightarrow 1 - \alpha$$

CONFIDENCE INTERVALS: INTERPRETATION

Interpreting a 95 % confidence interval:

- **Correct:** through repeated samples, 95 out of 100 confidence intervals would be expected to contain the true value of the parameter (with $\alpha = 0.05$)
- **Wrong:** through one sample, there is a 95 % chance the confidence interval around my $\hat{\theta}$ contains the true parameter θ (with $\alpha = 0.05$).

CONFIDENCE INTERVALS: INTERPRETATION

```
R <- 10000
n <- 25
set.seed(123456)
xbar <- replicate(R, mean(rnorm(n)))
ci.lo <- xbar - qnorm(.975) / sqrt(n)      # 95% CI lower bound
ci.up <- xbar + qnorm(.025) / sqrt(n)     # 95% CI upper bound
ci.in <- (ci.lo <= 0) & (0 <= ci.up)
summary(ci.in)
```

```
##      Mode    FALSE    TRUE
## logical    479    9521
```

```
mean(ci.in)
```

```
## [1] 0.9521
```

CHARACTERISTICS OF CONFIDENCE INTERVALS

Width and shape

The width of the confidence interval refers to the distance between the upper and lower bounds, i.e., $\text{width} = b(\hat{\theta}) - a(\hat{\theta})$

The shape of the confidence interval refers to the ratio of the distance between the upper bound and the estimate relative to the distance between the estimate and the lower bound, i.e., $\text{shape} = \frac{b(\hat{\theta}) - \hat{\theta}}{\hat{\theta} - a(\hat{\theta})}$. If $\text{shape} > 1$, this indicates that the confidence interval is wider on the right side, and if $\text{shape} < 1$, this indicates that the confidence interval is wider on the left side. Note that if $\text{shape} = 1$, then the confidence interval is symmetric around the estimate $\hat{\theta}$, which is typically the case for confidence intervals of mean parameters.

CHARACTERISTICS OF CONFIDENCE INTERVALS

Given a procedure for forming confidence interval, the **coverage rate** refers to the proportion of times that the parameter θ is included within the interval using some specified number of replications R . For example, the observed coverage rate in the sample mean example in slide 10 was 0.9521 using $R = 10000$ replications. If the coverage rate is above the nominal level of $1 - \alpha$, then the confidence interval procedure is said to be conservative. In contrast, if the coverage rate is below the nominal level, then the confidence interval procedure is said to be liberal. Finally, the confidence interval procedure is said to be “accurate” if the coverage rate is approximately equal to the nominal rate.

CHARACTERISTICS OF CONFIDENCE INTERVALS

Different confidence interval methods can be classified into two general types, which are distinguished by their level of accuracy. A **confidence interval procedure is first-order accurate** if the non-coverage probability on each side is such as:

$$P(\theta < a(\hat{\theta})) = \frac{\alpha}{2} + \frac{h_a}{\sqrt{n}} \quad \text{and} \quad P(\theta > b(\hat{\theta})) = \frac{\alpha}{2} + \frac{h_b}{\sqrt{n}}$$

where h_a and h_b are constants. A **confidence interval procedure is second-order accurate** if the non-coverage probability on each side is such as:

$$P(\theta < a(\hat{\theta})) = \frac{\alpha}{2} + \frac{h_a}{n} \quad \text{and} \quad P(\theta > b(\hat{\theta})) = \frac{\alpha}{2} + \frac{h_b}{n}$$

where h_a and h_b are constants.

CHARACTERISTICS OF CONFIDENCE INTERVALS.

IMPORTANT REMARKS

IMPORTANT: The purpose of any method for forming a confidence interval is to produce endpoints $a(\hat{\theta})$ and $b(\hat{\theta})$ that provide accurate coverage rates. Thus, the coverage rate is the most important aspect of any procedure for forming a confidence interval. If the coverage rate is not accurate, then the width and shape do not really matter. So it is only typical to discuss the width and shape of the confidence interval after ensuring that the **confidence interval procedure** results in accurate coverage rates.

CHARACTERISTICS OF CONFIDENCE INTERVALS.

IMPORTANT REMARKS

IMPORTANT: second-order accurate confidence intervals are preferred over first-order accurate intervals, given that the coverage rate converges to the nominal rate faster as $n \rightarrow \infty$. However, for any given sample of data, there is no guarantee which confidence interval method will perform best. In the long-run, the second-order accurate intervals should produce better results, but this does not guarantee that they will produce better results for every sample of data. Thus, in practice, it may be useful to compare **various different confidence interval** methods to get a sense of the uncertainty in the parameter estimate.

CHARACTERISTICS OF CONFIDENCE INTERVALS

Transformation invariance

Transformation invariance is a desirable property for confidence intervals. If h is a monotone transformation and $\phi = h(\theta)$, an interval is transformation invariant if the endpoints of the confidence interval for ϕ are $h(L)$ and $h(U)$ where (L, U) is the interval for θ .

If one person computes a confidence interval for θ and another one builds a CI for $\log(\theta)$ they get equivalent answers. For instance, if the interval for θ is $(1.31, 3.70)$ then, the interval for $\log(\theta)$ would be $(\log(1.31), \log(3.70)) = (0.273, 1.31)$.

SOME OBSERVATIONS

- The known confidence interval for a mean using the t distribution relies on the normal distribution of the population or on an enough large sample size.
- Confidence intervals for a mean using the t distribution are robust against most violations of the normal population assumption. However, you should check the data graphically to identify skewness or outliers that could affect the validity of the mean or its confidence interval.
- Also, unless the data production used randomization (random sample or a randomized experiment), statistical inference may be inappropriate.

OTHER ESTIMATION METHODS: THE BOOTSTRAP

The main benefit of the bootstrap is that it allows statisticians to set confidence intervals on a variety of parameters without having to make unreasonable assumptions. Let θ be the parameter we are interesting in estimating.

$$\theta = T(X_1, \dots, X_n)$$

The bootstrap idea is elegantly simple. First, you gather the sample and use it to find $\hat{\theta}$, your estimate of the unknown parameter θ .

Then draw a new random sample of size n , with replacement, from the original sample. This second sample is called a bootstrap sample. For that bootstrap sample, we can calculate an estimate of the parameter of interest, $\hat{\theta}_1^*$.

Then, we can draw as many bootstrap samples of size n as we want, obtaining B estimates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$.

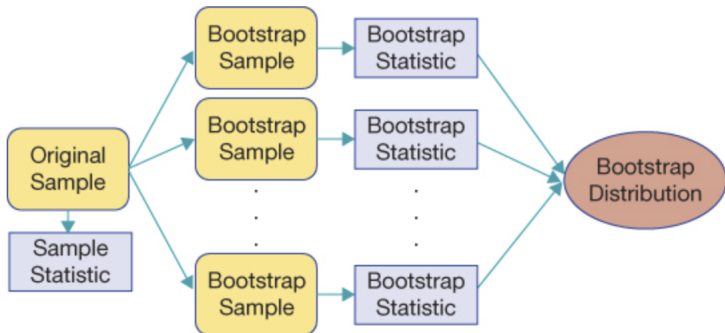
THE BOOTSTRAP

The bootstrap idea is illustrated in the diagram below ...

$$\begin{array}{ccccc} f & \longrightarrow & \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} & \Longrightarrow & \hat{f} \longrightarrow \begin{bmatrix} X_{1(\mathbf{1})}^* \\ X_{2(\mathbf{1})}^* \\ \vdots \\ X_{n(\mathbf{1})}^* \end{bmatrix}, \dots, \begin{bmatrix} X_{1(\mathbf{M})}^* \\ X_{2(\mathbf{M})}^* \\ \vdots \\ X_{n(\mathbf{M})}^* \end{bmatrix} \\ & & \downarrow & & \downarrow \qquad \qquad \qquad \downarrow \\ & & \hat{\theta} & & \hat{\theta}_{\mathbf{1}}^*, \dots, \hat{\theta}_{\mathbf{M}}^* \end{array}$$

THE BOOTSTRAP

... and also in the following diagram:



THE BOOTSTRAP IDEA

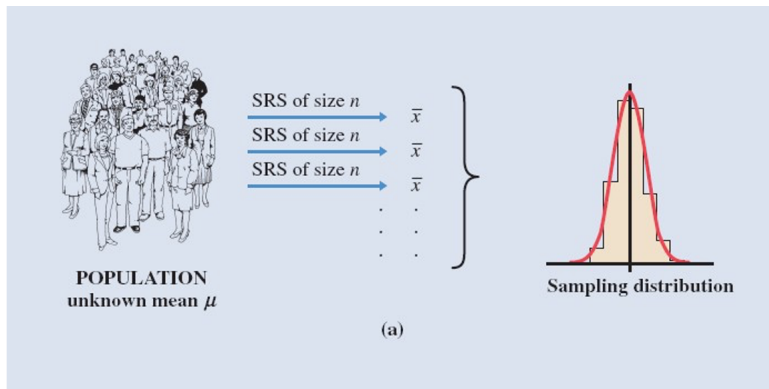
The original sample represents the population from which it was drawn. So resamples from this sample represent what we would get if we took many samples from the population. The bootstrap distribution of a statistic, based on many resamples, represents the sampling distribution of the statistic, based on many samples.

PROCEDURE FOR BOOTSTRAPPING

- Step 1: Resample. Create hundreds of new samples by sampling with replacement from the original random sample.
- Step 2: Calculate the bootstrap distribution. Calculate the statistics for each resample. The distributions of these statistics is called a bootstrap distribution.
- Step 3: Use the bootstrap distribution. The bootstrap distribution gives information about the shape and spread of the sampling distribution of the statistic.

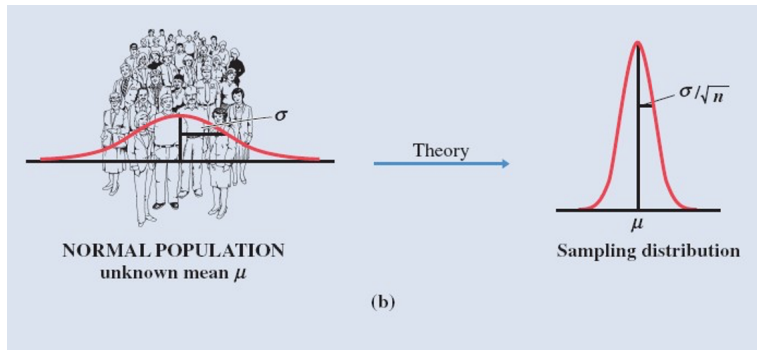
THEORY VS. SAMPLING

Sampling distribution of the sample mean (or any other statistic). Take many samples, collect the statistic values from each and look at the distribution of these values.



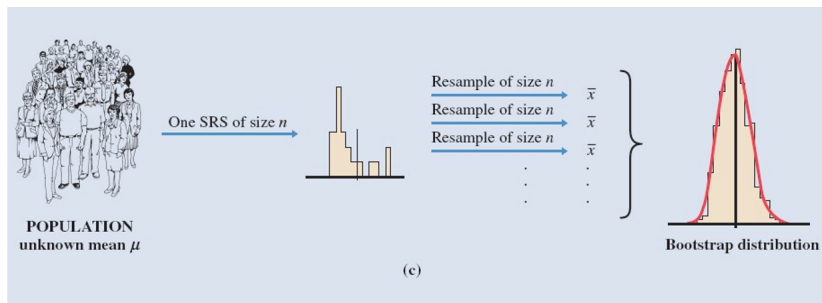
THEORY VS. SAMPLING

The probability theory shortcut: if we know that the population values follow a Normal distribution, theory tells us that the sampling distribution of $\bar{X}$ is also Normal.



THEORY VS. SAMPLING

The bootstrap idea: when theory fails and we can afford only one sample, that sample stands in for the population and the distribution of the statistic in many resamples stands in for the sampling distribution.



BOOTSTRAP DISTRIBUTIONS: BIAS AND STANDARD ERRORS

Bias:

- A statistic used to estimate a parameter is **biased** when its sampling distribution is not centered at the true value of the parameter. The bias of a statistic is the mean of the sampling distribution minus the parameter.

$$Bias = E[\hat{\theta}] - \theta$$

- The bootstrap method allows us to check for bias by seeing whether the bootstrap distribution of a statistic is centered at the statistic of the original random sample. **The bootstrap estimate of the bias is the mean of the bootstrap distribution minus the statistic for the original data, i.e., $\bar{\hat{\theta}}^* - \hat{\theta}$.**

BOOTSTRAP DISTRIBUTIONS: BIAS AND STANDARD ERRORS

Bootstrap Standard Error:

- The bootstrap standard error of a statistic is the standard deviation of the bootstrap distribution of that statistic.
- The bootstrap standard error based on B resamples is:

$$SE_{boot, \hat{\theta}} = \sqrt{\frac{1}{B-1} \sum_{i=1}^B \left(\hat{\theta}_i^* - \frac{1}{B} \sum_{i=1}^B \hat{\theta}_i^* \right)^2}$$

WHY DOES BOOTSTRAPPING WORKS?

- We compute a standard error by using resampling rather than the formula $\frac{s}{\sqrt{n}}$ (which is the standard error for the **mean**).
- We use the bootstrap distribution to see whether the sampling distribution is approximately Normal, rather than just hoping that our sample is large enough for the central limit theorem to apply.

Moreover:

- For most statistics, almost all the variation in bootstrap distributions comes from the selection of the original sample from the population. You can reduce this variation by using a larger original sample.
- The bootstrap resampling process using 1000 or more resamples introduces very little additional variation.
- Bootstrapping does not overcome the weakness of small samples as a basis for inference. Use caution in any inference—including bootstrap inference—from a small sample.

BOOTSTRAP CONFIDENCE INTERVALS

From that bootstrap distribution we can construct confidence intervals.
There are different methods:

- Interval using a bootstrap standard error
- Bootstrap percentile interval
- Reverse bootstrap percentile interval (not recommended!)
- Bootstrap t interval (bootstrap within bootstrap or studentized t-intervals or nested bootstrap)
- Bootstrap bias-corrected accelerated (BCa) confidence interval

INTERVAL USING A BOOTSTRAP STANDARD ERROR

Suppose that the bootstrap distribution of a statistic from a simple random sample (SRS) of size n is approximately Normal and that the bias is small. An approximate $(1 - \alpha)100\%$ confidence interval for the parameter that corresponds to this statistic by the plug-in principle is:

$$\hat{\theta} \pm t_{n-1, \alpha/2} SE_{boot}$$

where SE_{boot} is the bootstrap standard error for this statistic and $t_{n-1, \alpha/2}$ is the corresponding critical value of the t distribution with $n - 1$ degrees of freedom.

INTERVAL USING A BOOTSTRAP STANDARD ERROR

This interval doesn't have any real benefit over using the classic confidence interval formula.

Pros:

- Simple to form and easy to understand
- Can be applied when the standard error of the statistic is difficult to derive

Cons:

- Only first-order accurate
- Can perform poorly if the sampling distribution of $\hat{\theta}$ is highly skewed

BOOTSTRAP PERCENTILE INTERVAL

The interval between the $Q_{\alpha/2}$ and $Q_{1-\alpha/2}$ percentiles of the bootstrap distribution of a statistic is a $(1 - \alpha)100\%$ bootstrap percentile interval for the corresponding statistic. Use this method when the bootstrap estimate of bias is small. This method doesn't ignore skewness.

BOOTSTRAP PERCENTILE INTERVAL

Pros:

- Simple to form and easy to understand
- **Transformation Invariant**
- Advantage over t CI with bootstrap standard error when data are skewed

Cons:

- It only does partial skewness correction, which adds random variability
- Only first-order accurate

REVERSE BOOTSTRAP PERCENTILE INTERVAL (A.K.A. BASIC)

The reverse bootstrap interval argues that the behavior of $\hat{\theta} - \theta$ is approximately the same as the behavior of $\hat{\theta}^* - \hat{\theta}$. Then:

$$\begin{aligned}1 - \alpha &\approx P[Q_{\alpha/2} \leq \hat{\theta}^* \leq Q_{1-\alpha/2}] \\&= P[Q_{\alpha/2} - \hat{\theta} \leq \hat{\theta}^* - \hat{\theta} \leq Q_{1-\alpha/2} - \hat{\theta}] \\&\approx P[Q_{\alpha/2} - \hat{\theta} \leq \hat{\theta} - \theta \leq Q_{1-\alpha/2} - \hat{\theta}] \\&= P[Q_{\alpha/2} - 2 \cdot \hat{\theta} \leq -\theta \leq Q_{1-\alpha/2} - 2 \cdot \hat{\theta}] \\&= P[2 \cdot \hat{\theta} - Q_{1-\alpha/2} \leq \theta \leq 2 \cdot \hat{\theta} - Q_{\alpha/2}]\end{aligned}$$

REVERSE BOOTSTRAP PERCENTILE INTERVAL

The interval is then $(2 \cdot \hat{\theta} - Q_{1-\alpha/2}, 2 \cdot \hat{\theta} - Q_{\alpha/2})$. Assume that $Q_{\alpha/2} = \hat{\theta} - a$ and $Q_{1-\alpha/2} = \hat{\theta} + b$. Substituting, we find that the reverse bootstrap percentile interval is:

$$(\hat{\theta} - b, \hat{\theta} + a)$$

which is the mirror image of the bootstrap percentile interval. Be careful, **It is asymmetrical in the wrong direction!**

STUDENTIZED t INTERVALS

Each bootstrap sample is converted into a t-score as follows:

$$t_i^* = \frac{\hat{\theta}_i^* - \hat{\theta}}{SE_{\hat{\theta}_i^*}}$$

where $\hat{\theta}$ is the estimate from the original sample (non-bootstrap sample) and $\hat{\theta}_i^*$ is the statistics estimated from a single bootstrap sample. For $SE_{\hat{\theta}_i^*}$ we use a nested bootstrap algorithm:

- Draw a bootstrap sample and estimate the statistic $\hat{\theta}_i^*$.
- Bootstrap the bootstrap sample, and use the standard deviation of the estimate across bootstrapped samples as the estimate of $SE_{\hat{\theta}_i^*}$.
- Save $\hat{\theta}_i^*$ and $SE_{\hat{\theta}_i^*}$ and, for each bootstrap sample estimate t_i as above, $i = 1, \dots, B$.

STUDENTIZED t INTERVALS

Compute the $(1 - \alpha)100\%$ studentized t intervals as:

$$(\hat{\theta} - q_{1-\alpha/2} \cdot SE_{\hat{\theta}}, \hat{\theta} - q_{\alpha/2} \cdot SE_{\hat{\theta}})$$

where $\hat{\theta}$ is the sample statistic (without bootstrap); $SE_{\hat{\theta}}$ is the bootstrap estimate of the standard error (the standard deviation of bootstrap distribution; not the nested bootstrap part) and $q_{1-\alpha/2}$ and $q_{\alpha/2}$ are the percentiles of our nested bootstrap t -estimates $\{t_i^*\}$.

STUDENTIZED t INTERVALS

Pros:

- Simple idea with intuitive procedure
- Works well for location parameters
- Second-order accurate

Cons:

- Not transformation invariant
- Requires iterated bootstrap (computationally costly)
- Doesn't tend to work as well for correlation/association measures

BOOTSTRAP BCA CONFIDENCE INTERVAL

Bootstrap bias corrected and accelerated intervals use percentiles of the bootstrap distribution, but they don't necessarily use the $100\frac{\alpha}{2}$ -th and $100(1 - \frac{\alpha}{2})$ -th.

- Depend on acceleration parameter $\hat{a}$
- Depend on a bias-correction factor $\hat{z}_0$

BCa intervals have the form $(Q_{\alpha_1}, Q_{\alpha_2})$

- $\alpha_1 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{\frac{\alpha}{2}}}{1 - \hat{a}(\hat{z}_0 + z_{\frac{\alpha}{2}})}\right)$
- $\alpha_2 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{1 - \frac{\alpha}{2}}}{1 - \hat{a}(\hat{z}_0 + z_{1 - \frac{\alpha}{2}})}\right)$
- Φ is the cdf of the normal distribution
- z_{α} is the 100α -th percentile of the normal distribution.
- If $\hat{z}_0 = \hat{a} = 0$ then $\alpha_1 = \Phi(z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$ and $\alpha_2 = \Phi(z_{1 - \frac{\alpha}{2}}) = 1 - \frac{\alpha}{2}$, the BCa is the same as the percentile interval.

BOOTSTRAP BCA CONFIDENCE INTERVAL

- The bias-correction factor $\hat{z}_0$ measures median bias of $\hat{\theta}^*$, i.e., difference between $median(\hat{\theta}_b^*)$ and $\hat{\theta}$.
- The acceleration factor, $\hat{a}$, estimates the rate of change of the standard error of $\hat{\theta}$ with respect to the true parameter θ .

BOOTSTRAP BCA CONFIDENCE INTERVAL

Pros:

- Transformation invariant
- Works well for a variety of parameters
- Second-order accurate

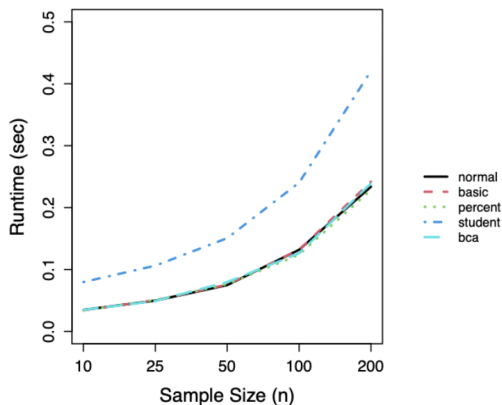
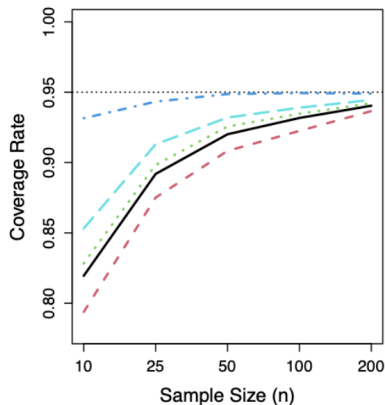
Cons:

- Less intuitive than other methods
- Requires estimation of acceleration and bias-correction

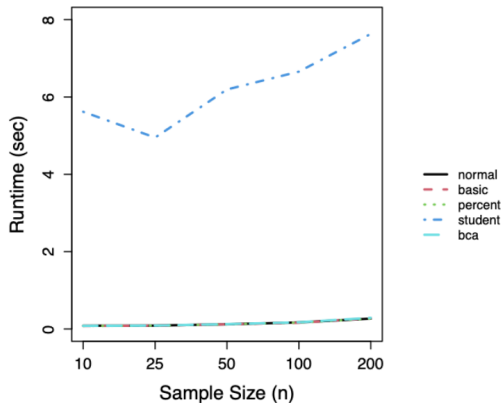
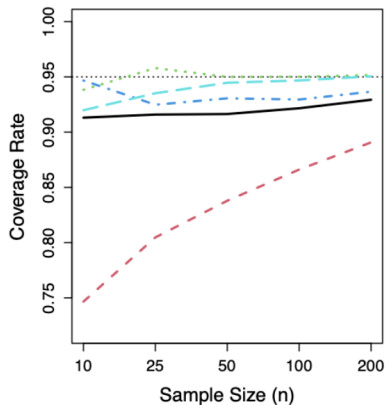
Some remarks:

- Bootstrap t and BCa confidence intervals are the best (generally speaking), specially when:
 - The statistic is strongly biased, as indicated by the bootstrap estimation of bias.
 - The sampling distribution of the statistic is clearly skewed, as indicated by the bootstrap distribution and by comparing the t and percentile intervals.
- Avoid bootstrapping the median and quantiles unless your sample is rather large!

COMPARING METHODS FOR A MEAN



COMPARING METHODS FOR A MEDIAN



THE BOOTSTRAP

Before the bootstrap, statisticians had to write all estimates as special kinds of averages and use the Central Limit Theorem to set approximate confidence intervals. But one can show that, as n gets large, the bootstrap is never worse than the Central Limit Theorem approximation and for many parameters can do much better.

THE BOOTSTRAP: SUMMARY

The bootstrap is a simulation method that resamples from the observed data. It treats the data distribution as if it were the population distribution.

Steps to use the bootstrap method, generating the bootstrap distribution:

- Generate bootstrap samples by sampling with replacement from the original sample, using the same sample size.
- Compute the statistic of interest, called a bootstrap statistic, for each of the bootstrap samples.
- Collect the statistics for many bootstrap samples to create a bootstrap distribution.